四川大学期末考试试题（闭卷）答案

（2024—2025 学年第 1 学期）　A 卷

课程号：304032030　课序号：1-8　课程名称：计算机网络（双语）　成绩：
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适用专业年级：2022　学生人数：　印题份数：　学号：　姓名：

请确保所有答案都书写在指定的答题纸上。

同时，请在试卷上的"考生承诺"区域签名，并记得将签名后的试卷与答题纸一并上交。

1.Acronym Match (10 points, 1 point for each acronym)

Consider the following acronyms: AIMD, ARP, CIDR, CSMA，DHCP, DNS, DSL, eBGP ,ECN, FDM, FIN, HOL, HTTP, iBGP，ICMP, LAN, MTU, NAT, RFC, RIP, SDN, SYN, TCP, TDM, TLD, UDP, URL, UTP。 Match the acronyms to the descriptions below, using each acronym exactly once. Supply exactly one answer for each item below. In some cases, an acronym might fit more than one question, but there is only one way to answer these so that every acronym is used exactly once.

1） The most used unreliable transport control protocol. UDP

2） A mail access protocol which has more features than POP3. HTTP

3） A algorithm is used wildly in IP routing to exchange routing information and spans two Autonomous Systems (AS) in the Internet. eBGP

4）A standardized address format used to specify the location of a web resource on the Internet. URL

5） A network protocol is typically used for diagnostic or control purpose. ICMP

6） The name of the message used to Close a TCP connection. FIN

7） An in-line network device that rewrites IP addresses (and often transport ports). NAT

8）. A wire pair constitutes a single communication link, and common used for computer network within a building. DSL 或 UTP

9）. A technique that allows multiple signals to be transmitted by dividing the bandwidth into many pieces. FDM

10）. As human, we have a protocol that allow us not only to behave with more civility, and to handle random access and decreases collision . CSMA

2. Single Choice: 15 Point , 1 points each question

1). In the internet, the equivalent concept to the end system is (*C*):

A. Client B. Server C. Host D. Switcher

2）. The following technologies may be used for residential access except (D)

A. HFC B. DSL C. Dial-Up modem D. FDDI

3）. The time required to propagate from the beginning of link to the next router is (D)

A. Processing delay B. queuing delay C. transmission delay

D. Propagation delay

4）. In the following descriptions about HTTP ,which is false (A):

A.HTTP use non-persistent connections in its default mode

B. HTTP use TCP as transport protocol.

C. HTTP is stateless protocol;.

D. HTTP is C/S architecture.

5）. In the following applications, which uses UDP ? (C)

A. Web B. file transfer C. DNS D. E-mail .

6）. Which is not provided by DNS (D)

A. Host aliasing B. mail-server aliasing C. Load distribution D. IP address allocation .

7）. The send side of transport layer fragments pass to (C) layer?

A. Application B. transport C. network D. link .

8） When you send a Email to QQ.com, the system use _（ 优选 D *或 A* ） protocol:

A) HTTP B) IMAP C) POP D) SMTP

9) TCP is a (B) protocol:

A) connection-oriented unreliable B) connection-oriented reliable

C) connectionless reliable D) connectionless unreliable

10). OSPF is a kind of (B) algorithm.

A) Distance Vector B) Link State C) Both of A,B D) Neither A and B

11) Which of following IP address doesn't belong to the 192.168.1.2/9 subnet? （B）

A) 192.178.1.1 B) 192.118.2.2 C) 192.222.3.4. D)192.234.5.6

12)There is a 6-bit data 101010, and the CRC is applied to it with generator 1001.Thus the CRC bit should be （ D ）

A) 100 B) 001 C) 110 D) 111

13). ICMP is used for (D).

A.RDT B.Flow control C E-mail D. Error report

14). For（ B ）Link that have a Single sender at one end of the link and many receivers at the other end of the link.

A. Switch B. Broadcast C. Multicast D. All of A,B C

15).In an Ethernet frame, the preamble is responsible for （ D ）

A) Error detection B) Collision Detection

C) Multiplexing D) Synchronization of the receiver to sender.

3. Decide True or False. if false, give out the correct answer. (10 Points , 2 points each question)

1) Suppose Host A is sending a large file to Host B over a TCP connection. If the sequence number for a segment of this connection is **m**，then the sequence number for the subsequent segment will necessarily be **m+1**.

False.

与发送了 **segment** 的长度有关， 每 **Byte** 为 **1** 段号。根据实际长度 增加。

2) Consider TCP congestion control, when the timer expired at the sender(congestion occurred), the value of **ssthresh** is cut to half of its previous **ssthresh.**

False,

新 **ssthresh** 为发生 **Congestion** 处值的一半。

3）. Queuing is always occurred only in the input of router.

False。

Input and output 都会有 **Queuing** 发生。

4） Forwarding table is configured only by intra-AS routing algorithm, not by inter-AS routing algorithm.

False。

Forwarding table 在 Inter-AS, Intra-AS 都要配置。

5） Among multiple access methods, both FDM and TDM can avoid collision.

True

4. Discribe and answer the following Questions: 20 Point , 5 points each question

1） In our Top-Down OSI Model, what are layers in the internet protocol stack? What are the basic functions of each of these layers.

Five layers： Application，Transport, network, link, physical

5 分、每层 1 分。功能不完整；适当扣 0.5 分

Application: 各种运用， 如 HTTP， email

Transport, RDT , TCP/UDP

network, routing, switch

link, access Link， frame，error detection, access 等

physical: 提供物理资源， 介质。 如有线，无线、光纤等

2） In the TCP connection manage , special process and special segment are required. Please give out how a TCP connection is established. Suppose client host A， server host B.

(1) step 1: 特殊 SYN=1， 配置 Seq 号。发起端发送

(2) step 2 服务器端收到， 回复 SYN=1， 配置 Seq 号、ACK。

(3) step 3 回复 SYN=0， 配置 Seq 号、ACK。

可以给出图， 每不 1 分，共 3 分。 SYN+Seg# +ACK# 2 分。

不完整，适当扣分。

3） In Version 7 book, the network lay functionality is divided into data plane functionality and control plane functionality. What are the main functions of data plane?

main functions of data plane：key point 每点 1 分。

（1） 侧重 per-router, R 的组成、实现

（2） forwarding table 的产生。

（3） switch

（4） IP datagram format

(5) data deliver

4） Random access CSMA /CD is always used in LAN, give out the processes and methods to avoid collision.

每要点 各 1 分。

(1) Listen: before sending

(2) if busy, no sending

(3) if idle, no collision, sending.
(4) collision,jam, backoff
(5) once more, from (1)

5. Integrated Question: 45 Point , 15 points each question.

5-1 Suppose two hosts, A and B, are separated by 20,000 kilometers and are connected by a direct link of R= 2 Mbps. Suppose the propagation speedover the link is $2.5x10^8$ meters/sec.

1）. Dprop= 20000 Km÷ $2.5x10^8$ == $8*10^{-2}$ S**(2 Point)**

R*dprop. = 2 Mbps* $8*10^{-2}$ S=160000 bits **(3 Point)**

2） 160000 bits **(5 Point)**

3). 能够传输的最多数据. **(5 Point)**

5-2 Suppose an ISP gives us a block of address 192.168.40.128/25. Please create four class-subnets A, B, C ,D. A has 50 students， B has 25 students， C has 10 students，D has 10 students: **(15 Point)**

a) **(8 point, 2 points each)** 可能有多重答案

7bit: x1 x x x x x x 下列为： A X1=0

B: 1 0 xxxxx

C: 1 1 XXXxx

D 1 1 1 xxxx

A 192.168.40.128/26 B 192.168.40.192/27

C 192.168.40.224/28 D 192.168.40.240/28

b) A 255.255.255.192 **(5 pointS)**

B 255.255.255.224 C 255.255.255.240 D 255.255.255.240

b) A 192.168.40.191 **(2 point)**

B 192.168.40.223 C 192.168.40.239 B 192.168.40.255

5-3 Consider the following network. With the indicated link costs, use Dijkstra's shortest-path algorithm to compute the shortest path from **x** to all network nodes.

1).Show how the algorithm works by computing a table as following **(10 Point)**

下列过程可以简化　　**(修改为 10 Point)**

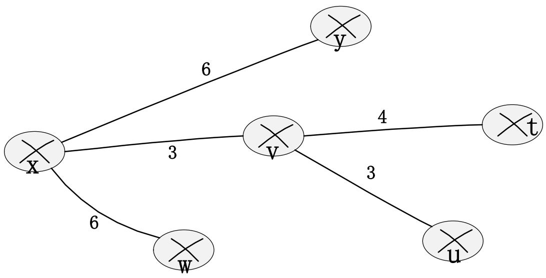
Step	N’	D(u),p(u)	D (v),p(v)	D(w),p(w)	D(y),p(y)	D(t),p(t)
0	x	∝, v	3, x	6, x	6,x	∝,
1	x v	6，v		6，x	6,x	7, v
2	x v u			6,x	6,x	7,v
3	x v u w				6,x	7,v
4	x v u w y				6,x	7,v
5	x v u w y t	……….				7,v

2) give out the shortest path from **x** to all other node.　**(修改为 3 Point)**

也可以文字说明、　或给表（横、竖　皆可以）

X to	u	v	w	y	t	x
shortest path	6	3	6	6	7	0

只给出下列图也可



3) give out the forwarding table for **x.**　**(修改为 2 Point)**

表（横、竖　皆可以）

Destination	u	v	w	y	t
Link	（x,v）	(x,v)	(x,w)	(x,y)	(x,v)